

SNA Fall 2025: Assignment 5

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Libraries and Data

Libraries

```
library(sna)
library(statnet)
library(igraph)
library(ergm)
library(network)
```

Data

```
davis_df <- read.csv("davis.csv", header = TRUE, check.names = FALSE)
```

Explore

```
class(davis_df)
```

```
## [1] "data.frame"
```

```
dim(davis_df) # 89 rows and 2 columns
```

```
## [1] 89 2
```

```

head(davis_df)

##           w e
## 1 EVELYN 1
## 2 EVELYN 2
## 3 EVELYN 3
## 4 EVELYN 4
## 5 EVELYN 5
## 6 EVELYN 6

g_davis <- graph_from_data_frame(davis_df)

davis <- upgrade_graph(g_davis)
summary(davis)

## IGRAPH 6bfda93 DN-- 32 89 --
## + attr: name (v/c)

events <- as.character(davis_df$e)
V(davis)$type <- V(davis)$name %in% events # events = TRUE, woman = FALSE

```

Task 1: Network Description

Study Purpose and Context

The data originates from a comparative ethnography of social class in Natchez, Mississippi, conducted in the 1930s by Davis, Gardner, and Gardner (DGG). The researchers aimed to examine the correspondence between social class levels and patterns of informal interaction. Specifically, they sought to determine the extent to which informal contacts were established within an individual's own class level (Freeman, L. C., 2003, p.3)

To achieve this, they collected systematic data on the social activities of 18 women over a nine-month period, observing which subsets of women met across a series of 14 informal social events (Freeman, L. C., 2003, p.4). Their primary goals were to divide the women into distinct “cliques” (groups) based on co-attendance and to determine the specific position (core vs. peripheral) of each woman within those groups (Freeman, L. C., 2003, p.6).

The Data Source

The network consists of a “two-mode” dataset (woman-by-event), which captures the participation of the 18 women in the 14 events. The data were originally compiled from interviews, the records of participant observers, guest lists, and newspapers (Freeman, L. C., 2003, p.4).

Breiger's Matrix Algebraic Analysis

Breiger (1974) utilized matrix algebra to analyze the network, highlighting the mathematical “duality” of the two-mode data. By multiplying the original matrix by its transpose, he generated two distinct one-mode matrices (Freeman, L. C., 2003, p.14):

- Woman-by-Woman: Measuring the number of events co-attended by pairs of women.
- Event-by-Event: Measuring the number of women who attended both events in a pair.

Key Info about the Networks

```

vcount(davis)

## [1] 32

```

```

ecount(davis)

## [1] 89

is.directed(davis)

## [1] TRUE

list.vertex.attributes(davis)

## [1] "name" "type"

list.edge.attributes(davis)

## character(0)

is_bipartite(davis)

## [1] TRUE

head(data.frame(name = V(davis)$name, type = V(davis)$type))

##      name  type
## 1  EVELYN FALSE
## 2   LAURA FALSE
## 3  THERESA FALSE
## 4   BRENDA FALSE
## 5 CHARLOTTE FALSE
## 6  FRANCES FALSE

table(vertex_attr(davis, "type"))

##
## FALSE  TRUE
##    18    14

```

Task 2: Construct, Visualize, and Describe the Two-Mode Network

2.1 Create Three Matrices

```

# matrix 1: woman by event
woman_by_event <- as_incidence_matrix(davis, types = V(davis)$type)

cat("Woman × Event Matrix:\n")

## Woman × Event Matrix:

print(woman_by_event)

##      1 2 3 4 5 6 8 9 7 12 10 13 14 11
## EVELYN 1 1 1 1 1 1 1 1 0 0 0 0 0 0
## LAURA  1 1 1 0 1 1 1 0 1 0 0 0 0 0
## THERESA 0 1 1 1 1 1 1 1 1 0 0 0 0 0
## BRENDA  1 0 1 1 1 1 1 0 1 0 0 0 0 0
## CHARLOTTE 0 0 1 1 1 0 0 0 1 0 0 0 0 0
## FRANCES 0 0 1 0 1 1 1 0 0 0 0 0 0 0
## ELEANOR 0 0 0 0 1 1 1 0 1 0 0 0 0 0
## PEARL   0 0 0 0 0 1 1 1 0 0 0 0 0 0
## RUTH    0 0 0 0 1 0 1 1 1 0 0 0 0 0

```

```
## VERNE      0 0 0 0 0 0 1 1 1 1 0 0 0 0
## MYRNA      0 0 0 0 0 0 1 1 0 1 1 0 0 0
## KATHERINE  0 0 0 0 0 0 1 1 0 1 1 1 1 0
## SYLVIA     0 0 0 0 0 0 1 1 1 1 1 1 1 0
## NORA       0 0 0 0 0 1 0 1 1 1 1 1 1 1
## HELEN      0 0 0 0 0 0 1 0 1 1 1 0 0 1
## DOROTHY    0 0 0 0 0 0 1 1 0 0 0 0 0 0
## OLIVIA     0 0 0 0 0 0 0 1 0 0 0 0 0 1
## FLORA      0 0 0 0 0 0 0 1 0 0 0 0 0 1
```

```
# matrix 2: woman by woman
```

```
woman_by_event_transpose <- t(woman_by_event)
```

```
woman_by_woman <- woman_by_event %*% woman_by_event_transpose
```

```
cat("Woman x Woman Matrix:\n")
```

```
## Woman x Woman Matrix:
```

```
print(woman_by_woman)
```

```
##          EVELYN LAURA THERESA BRENDA CHARLOTTE FRANCES ELEANOR PEARL RUTH
## EVELYN      8      6      7      6      3      4      3      3      3
## LAURA      6      7      6      6      3      4      4      2      3
## THERESA     7      6      8      6      4      4      4      3      4
## BRENDA      6      6      6      7      4      4      4      2      3
## CHARLOTTE   3      3      4      4      4      2      2      0      2
## FRANCES     4      4      4      4      2      4      3      2      2
## ELEANOR     3      4      4      4      2      3      4      2      3
## PEARL       3      2      3      2      0      2      2      3      2
## RUTH        3      3      4      3      2      2      3      2      4
## VERNE       2      2      3      2      1      1      2      2      3
## MYRNA       2      1      2      1      0      1      1      2      2
## KATHERINE   2      1      2      1      0      1      1      2      2
## SYLVIA      2      2      3      2      1      1      2      2      3
## NORA        2      2      3      2      1      1      2      2      2
## HELEN       1      2      2      2      1      1      2      1      2
## DOROTHY     2      1      2      1      0      1      1      2      2
## OLIVIA      1      0      1      0      0      0      0      1      1
## FLORA       1      0      1      0      0      0      0      1      1
##          VERNE MYRNA KATHERINE SYLVIA NORA HELEN DOROTHY OLIVIA FLORA
## EVELYN      2      2      2      2      2      1      2      1      1
## LAURA      2      1      1      2      2      2      1      0      0
## THERESA     3      2      2      3      3      2      2      1      1
## BRENDA      2      1      1      2      2      2      1      0      0
## CHARLOTTE   1      0      0      1      1      1      0      0      0
## FRANCES     1      1      1      1      1      1      1      0      0
## ELEANOR     2      1      1      2      2      2      1      0      0
## PEARL       2      2      2      2      2      1      2      1      1
## RUTH        3      2      2      3      2      2      2      1      1
## VERNE       4      3      3      4      3      3      2      1      1
## MYRNA       3      4      4      4      3      3      2      1      1
## KATHERINE   3      4      6      6      5      3      2      1      1
## SYLVIA      4      4      6      7      6      4      2      1      1
## NORA        3      3      5      6      8      4      1      2      2
## HELEN       3      3      3      4      4      5      1      1      1
```

```
## DOROTHY      2      2      2      2      1      1      2      1      1
## OLIVIA       1      1      1      1      2      1      1      2      2
## FLORA        1      1      1      1      2      1      1      2      2
```

```
# matrix 3: event by event
```

```
event_by_event <- woman_by_event_transpose %*% woman_by_event
```

```
cat("Event × Event Matrix:\n")
```

```
## Event × Event Matrix:
```

```
print(event_by_event)
```

```
##      1 2 3 4 5 6 8 9 7 12 10 13 14 11
## 1    3 2 3 2 3 3 3 1 2 0 0 0 0 0
## 2    2 3 3 2 3 3 3 2 2 0 0 0 0 0
## 3    3 3 6 4 6 5 5 2 4 0 0 0 0 0
## 4    2 2 4 4 4 3 3 2 3 0 0 0 0 0
## 5    3 3 6 4 8 6 7 3 6 0 0 0 0 0
## 6    3 3 5 3 6 8 7 4 5 1 1 1 1 1
## 8    3 3 5 3 7 7 14 9 8 5 4 2 2 1
## 9    1 2 2 2 3 4 9 12 5 5 4 3 3 3
## 7    2 2 4 3 6 5 8 5 10 4 3 2 2 2
## 12   0 0 0 0 0 1 5 5 4 6 5 3 3 2
## 10   0 0 0 0 0 1 4 4 3 5 5 3 3 2
## 13   0 0 0 0 0 1 2 3 2 3 3 3 3 1
## 14   0 0 0 0 0 1 2 3 2 3 3 3 3 1
## 11   0 0 0 0 0 1 1 3 2 2 2 1 1 4
```

2.2 Create Three Visualizations

```
# plot 1: "woman by event" network
```

```
set.seed(123)
```

```
custom_colors <- c(
  rep("red", nrow(woman_by_event)), # women
  rep("blue", ncol(woman_by_event)) # events
)
```

```
gplot(
  woman_by_event,
  gmode = "twomode",
  label = c(rownames(woman_by_event), colnames(woman_by_event)),
  mode = "fruchtermanreingold",
  vertex.size = 1.5,
  usearrows = FALSE,
  edge.col = "light grey",
  edge.width = 0.1,
  vertex.border = "black",
  label.col = custom_colors,
  label.cex = 0.7,
  main = "Fig 1. Two-Mode Woman x Event Network"
)
```

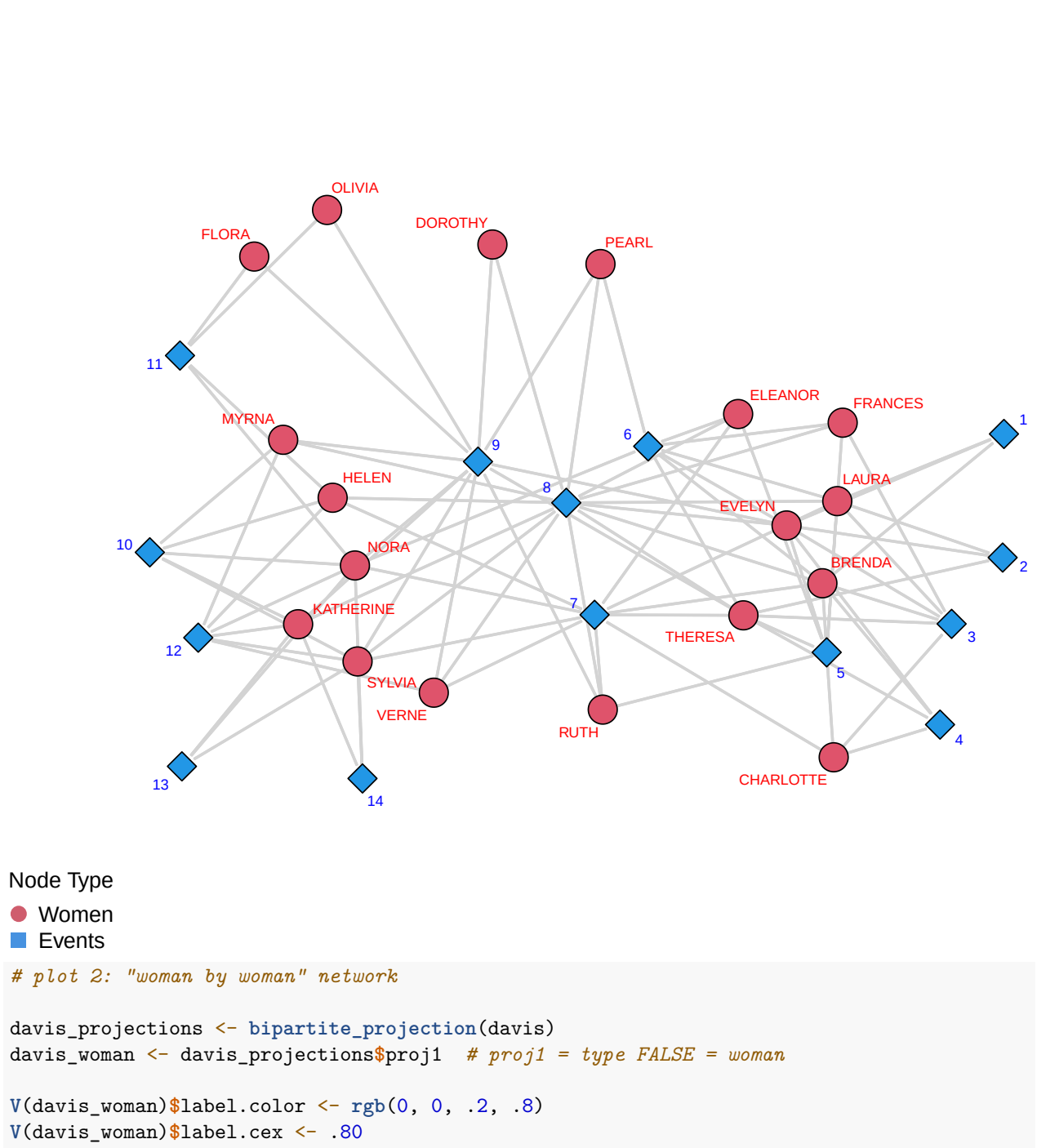
```
legend("bottomleft",
```

```

legend = c("Women", "Events"),
col = c("#CF5D6D", "#4792E0"),
pch = c(19, 15),
pt.cex = 1.5,
bty = "n",
title = "Node Type"
)

```

Fig 1. Two-Mode Woman x Event Network



```

V(davis_woman)$color <- rgb(0, 0, 1, .3)
V(davis_woman)$frame.color <- V(davis_woman)$color

egalpha <- log1p(E(davis_woman)$weight) / max(log1p(E(davis_woman)$weight) * 2)
E(davis_woman)$color <- rgb(.25, .75, 0, egalpha)

std_weight <- sd(E(davis_woman)$weight)
weight_mean_center <- (E(davis_woman)$weight - mean(E(davis_woman)$weight))

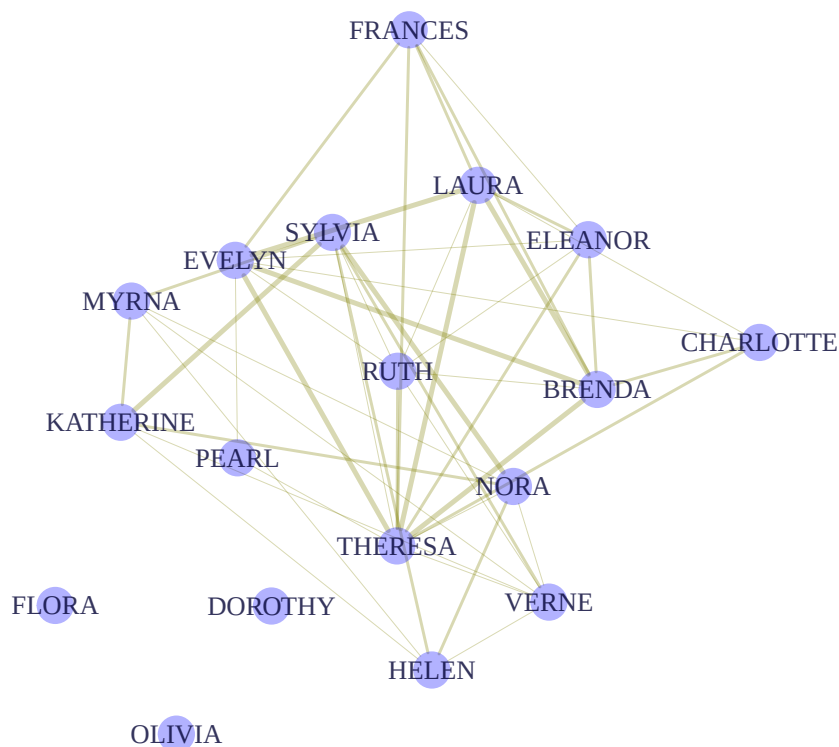
recode_weight <- E(davis_woman)$weight
recode_weight[weight_mean_center <= 0] <- -1
recode_weight[(weight_mean_center > 0) & (weight_mean_center <= std_weight)] <- .5
recode_weight[(weight_mean_center > std_weight) &
               (weight_mean_center <= std_weight * 2)] <- 1.5
recode_weight[weight_mean_center > std_weight * 2] <- 2.5

E(davis_woman)$color <- rgb(.5, .5, 0, .3)
E(davis_woman)$width <- recode_weight
V(davis_woman)$size <- 10

set.seed(123)
plot(davis_woman, layout = layout_with_lgl, main = "Fig 2. Woman x Woman Network")

```

Fig 2. Woman x Woman Network



```

# plot 3: "event by event" network

davis_event <- davis_projections$proj2 # proj2 = type TRUE = events

```

```

V(davis_event)$label.color <- rgb(0, 0, .2, .8)
V(davis_event)$label.cex <- 1.2
V(davis_event)$color <- rgb(0, 0, 1, .3)
V(davis_event)$frame.color <- V(davis_event)$color

egalpha <- log1p(E(davis_event)$weight) / max(log1p(E(davis_event)$weight) * 2)
E(davis_event)$color <- rgb(.25, .75, 0, egalpha)

std_weight <- sd(E(davis_event)$weight)
weight_mean_center <- (E(davis_event)$weight - mean(E(davis_event)$weight))

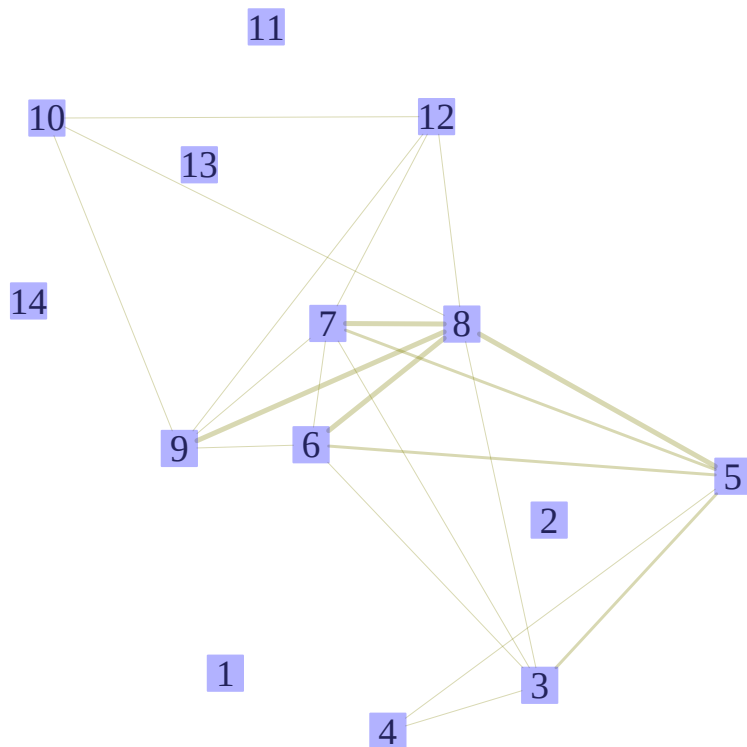
recode_weight <- E(davis_event)$weight
recode_weight[weight_mean_center <= 0] <- -1
recode_weight[(weight_mean_center > 0) & (weight_mean_center <= std_weight)] <- .5
recode_weight[(weight_mean_center > std_weight) &
               (weight_mean_center <= std_weight * 2)] <- 1.5
recode_weight[weight_mean_center > std_weight * 2] <- 2.5

E(davis_event)$color <- rgb(.5, .5, 0, .3)
E(davis_event)$width <- recode_weight
V(davis_event)$size <- 10
V(davis_event)$shape <- "square"

set.seed(123)
plot(davis_event, layout = layout_with_lgl, main = "Fig 3. Event × Event Network")

```

Fig 3. Event × Event Network



The original plot is pretty dense, making it harder to interpret. So I pare things down a bit, weighting the edges to produce a simpler plot. I will make the width of the edges proportional to the edge weight, based on the following logic.

I first calculate the mean and standard deviation of the raw weights. Then I assign a new weight, determined by how much the raw weight is below/above the mean. All edge weights equal to or below the mean get a -1 in my new weighting scheme, which will reduce the density of the plot considerably (I set the weight to -1 to ensure that these edges do not show up in the plot). All edges between 0 and 1 standard deviations above the mean weight get a 0.5, while all edges between 1 and 2 standard deviations get a 1.5. Finally, all edges above 2 standard deviations get a 2.5.

I will now apply my new weighting scheme to the widths of the edges. I will also change the color of the edges (to be the same across all edges) (Rawlings et al., 2023, Chapter 11.3.2).

The plot offers important insight into the structure of informal social grouping among the women. We can see that by filtering out edges with weights at or below the mean, the dense “ball of yarn” turns into a clear core-periphery structure.

For woman x woman network: there are a number of isolated or loosely connected members on the periphery, such as Olivia and Flora, and a core consisting of highly interconnected women, like Evelyn, Sylvia, and Theresa, who attended multiple events together .

For event x event network: there are a number of smaller, less overlapping events on the periphery (such as Event 1 or Event 11) and a core consisting of “bridging” events, like Event 6, 7, 8 and Event 9, which drew participants from across different social circles.

2.3 Describe

2.3.1 Describe the Networks you generated

First, for “woman x event” matrix, this is a raw “two-mode” matrix that records the direct participation of women in events. A tie (a value of 1) means a specific woman was present at a specific event and 0 indicates she was not present. For example, there is a 1 at the intersection of EVELYN and event 1, meaning Evelyn attended Event 1.

Second, for “woman x woman” matrix, this is a “one-mode projection” derived from the original data (calculated as $A \times A^T$). It represents the social network of the women, showing how connected they are based on shared activities(Freeman, L. C., 2003, p.14). A tie means the number of events two women attended together, and a higher number implies a stronger social bond. For example, the intersection of EVELYN and LAURA has a value of 6. This means Evelyn and Laura attended 6 events together, indicating a very strong relationship.

Third, for “event x event” matrix, this is the dual “one-mode projection” (calculated as $A^T \times A$). It represents the network of events, showing how similar or “overlapping” the events were in terms of their audience(Freeman, L. C., 2003, p.14). A tie means the number of women attended both events, and a higher number suggests the events drew from the same social circle. For example, the intersection of event 8 and event 9 has a value of 9. This means that 9 women attended both Event 8 and Event 9, suggesting these two events served nearly the exact same group of people.

Fourth, for “woman x woman” matrix, diagonal entries (e.g., EVELYN, EVELYN) tell us how many events each woman attended (her total activity). If `woman_by_woman["EVELYN", "EVELYN"] = 8`, Evelyn attended 8 events total. For “event x event” matrix, diagonal entries (e.g., 8, 8) tell us how many women attended each event (the popularity or size of the event). If `event_by_event["8", "8"] = 14`, 14 women attended Event 8.

Finally, Breiger’s concept of duality translates a classical sociological metaphor into matrix algebra to analyze how individuals and groups mutually define one another(Breiger, R. L., 1974, p.2). In other words, Breiger’s duality theory suggests that the relationship between persons and groups is mutually defined. Through

matrix operations, the original two-mode matrix can be transformed into two dual networks: an interpersonal network based on joint participation, and an event network based on common members (Breiger, R. L., 1974).

In this network (Davis Southern Women dataset), duality means that the social structure of the women is inextricably linked to, and mathematically defined by, the structure of the events they attended. Groups of women can be identified because they repeatedly attend the same subset of events, and, symmetrically, clusters of events can be identified because they consistently draw the same subset of women. Thus, the social grouping among women and the grouping among events are not independent findings but dual expressions of the same underlying pattern of affiliation. In Breiger’s terms, duality in the Southern Women network means that women’s groups and event clusters mutually define each other, so the same two-mode pattern simultaneously produces the group structure seen among women and its mirror image among events (Breiger, R. L., 1974).

2.3.2 Describe Two-Mode Networks more generally

In many cases, researchers choose to use Two-Mode network because of its advantages.

First, researchers often use Two-Mode networks when it is too difficult or sensitive to collect direct “person-to-person” data. This is very common in epidemiology (the study of diseases). As Frost (2007) explains, it is socially easier for a person to admit “I visited bar X” than “I had sexual contact with Person Y.” By linking people to the places they go (a “sexual affiliation network”), researchers can understand the potential spread of diseases like HIV without needing a complete list of direct partners. Frost notes that this method is often more robust and less biased than trying to trace direct contacts (Frost, 2007).

Second, researchers also use Two-Mode networks because converting data into a One-Mode network causes a loss of information. For example, if you only look at which women know each other, you lose the information about where they met. Borgatti and Everett (1997) argue that analyzing the Two-Mode data directly allows researchers to see the “duality” of social life. This means we can not only know that Evelyn and Laura are friends, but also know how they sustain that friendship (by attending the same events), as shown in the Davis Southern Women dataset.

Third, Two-Mode networks help researchers to understand some hidden connections or patterns. For example, Ahn et al. (2011) used a Two-Mode network to connect culinary ingredients (like shrimp or lemon) with the chemical flavor compounds they contain. This allowed them to discover the “food pairing hypothesis,” showing that Western cuisines prefer ingredients that share flavor compounds, while East Asian cuisines avoid them. Similarly, Muetzel and Breiger (2021) note that Two-Mode networks allow sociologists to analyze text by connecting documents to words. This helps reveal hidden patterns of meaning and culture that would be invisible in a regular One-Mode network.

However, it also has many limitations:

First, the most significant sociological limitation, noted by Frost (2007), is that co-presence does not guarantee interaction. Just because two people attended the same large event (e.g., a concert or a busy bar) does not mean they know each other or interacted. In a One-Mode projection, these two people would appear connected, potentially creating “false positives” in the social structure.

Second, in general, visualizing Two-Mode networks is difficult. As we can see, these graphs can become “ball of yarn” because highly popular events connect to many people, creating dense tangles of lines that are hard to interpret compared to simple One-Mode sociograms.

Third, standard network metrics like density and centrality require modification in Two-Mode networks, because connections within the same set of nodes are prohibited, complicating analysis (Borgatti & Everett, 1997, p.254).

But Two-mode networks still matter because they model the context of social relations. For example, in public health (Frost, 2007), knowing the places that bridge different risk groups is often more actionable than knowing the individuals. Public health officials can intervene at a specific bar or bathhouse to stop the spread of infection, effectively treating the “node” (venue) to protect the network. We rarely interact in a vacuum; we interact in focus circles, workplaces, and gathering spots. Two-mode networks treat these

“spots” as active parts of the social structure, not just background noise. In other words, two-mode networks are vital for uncovering the “mutual constitution” of actors and their environments, providing a rigorous method to study the intersection of individuals and social structure.

Task 3: Compute and Interpret Two-Mode Networks

3.1 “Woman by Event” Centrality Measure

```
# woman degree centrality
woman_degree <- degree(davis, v = V(davis)[type == FALSE])

cat("=== Woman Degree Centrality (Events Attended) ===\n")
```

```
## === Woman Degree Centrality (Events Attended) ===
```

```
print(woman_degree)
```

```
##      EVELYN      LAURA  THERESA  BRENDA CHARLOTTE  FRANCES  ELEANOR  PEARL
##          8          7          8          7          4          4          4          3
##      RUTH      VERNE      MYRNA KATHERINE  SYLVIA      NORA      HELEN  DOROTHY
##          4          4          4          6          7          8          5          2
##  OLIVIA      FLORA
##          2          2
```

```
mean(woman_degree)
```

```
## [1] 4.944444
```

```
V(davis)$degree <- degree(davis)
busy_idx <- V(davis)$type == FALSE

busy_woman <- data.frame(
  woman = V(davis)[busy_idx]$name,
  event = V(davis)[busy_idx]$degree
)
```

```
busy_woman[order(busy_woman$event, decreasing = TRUE), ]
```

```
##      woman event
## 1    EVELYN    8
## 3    THERESA    8
## 14   NORA      8
## 2    LAURA    7
## 4    BRENDA    7
## 13   SYLVIA    7
## 12 KATHERINE    6
## 15   HELEN     5
## 5    CHARLOTTE  4
## 6    FRANCES    4
## 7    ELEANOR    4
## 9     RUTH      4
## 10   VERNE      4
## 11   MYRNA      4
## 8     PEARL     3
## 16  DOROTHY     2
## 17  OLIVIA     2
```

```
## 18      FLORA      2
# event degree centrality
event_degree <- degree(davis, v = V(davis)[type == TRUE])

cat("=== Event Degree Centrality (Attendance Count) ===\n")

## === Event Degree Centrality (Attendance Count) ===
print(event_degree)

## 1  2  3  4  5  6  8  9  7 12 10 13 14 11
## 3  3  6  4  8  8 14 12 10  6  5  3  3  4

mean(event_degree)

## [1] 6.357143
V(davis)$degree <- degree(davis)
popular_idx <- V(davis)$type == TRUE

popular_event <- data.frame(
  event = V(davis)[popular_idx]$name,
  woman = V(davis)[popular_idx]$degree
)

popular_event[order(popular_event$woman, decreasing = TRUE), ]

##      event woman
## 7         8    14
## 8         9    12
## 9         7    10
## 5         5     8
## 6         6     8
## 3         3     6
## 10        12     6
## 11        10     5
## 4         4     4
## 14        11     4
## 1         1     3
## 2         2     3
## 12        13     3
## 13        14     3
```

Borgatti & Everett(1997) note that in a two-mode network:

- A woman can only participate in all the events (she cannot “connect” other women directly).
- An event can only be attended by all the women (it cannot be “connected” to other events directly).

Therefore, the normalization formula is “divide by the size of the opposite vertex set” and is usually expressed as a percentage:

- Normalized Degree for Women = $\frac{\text{Number of events the woman attended}}{\text{Total number of events (14)}} \times 100$
- Normalized Degree for Events = $\frac{\text{Number of women attending the event}}{\text{Total number of women (18)}} \times 100$

```
## Normalization
```

```
# raw degree centrality
```

```

V(davis)$degree <- degree(davis)

# count the size of each mode
num_women <- sum(V(davis)$type == FALSE)
num_events <- sum(V(davis)$type == TRUE)

# Borgatti & Everett 1997 normalization
V(davis)$norm_degree <- ifelse(V(davis)$type == FALSE,
                               (V(davis)$degree / num_events),
                               (V(davis)$degree / num_women))

# women normalized degree scores
women_scores <- data.frame(
  woman_name = V(davis)$name[V(davis)$type == FALSE],
  raw_degree = V(davis)$degree[V(davis)$type == FALSE],
  norm_degree = round(V(davis)$norm_degree[V(davis)$type == FALSE], 3)
)
women_scores <- women_scores[order(women_scores$norm_degree, decreasing = TRUE), ]

# event normalized degree scores
event_scores <- data.frame(
  event_name = V(davis)$name[V(davis)$type == TRUE],
  raw_degree = V(davis)$degree[V(davis)$type == TRUE],
  norm_degree = round(V(davis)$norm_degree[V(davis)$type == TRUE], 3)
)
event_scores <- event_scores[order(event_scores$norm_degree, decreasing = TRUE), ]

women_scores

```

```

##   woman_name raw_degree norm_degree
## 1    EVELYN          8      0.571
## 3   THERESA          8      0.571
## 14    NORA          8      0.571
## 2    LAURA          7      0.500
## 4    BRENDA          7      0.500
## 13   SYLVIA          7      0.500
## 12 KATHERINE          6      0.429
## 15    HELEN          5      0.357
## 5   CHARLOTTE          4      0.286
## 6    FRANCES          4      0.286
## 7    ELEANOR          4      0.286
## 9      RUTH          4      0.286
## 10   VERNE          4      0.286
## 11   MYRNA          4      0.286
## 8     PEARL          3      0.214
## 16  DOROTHY          2      0.143
## 17   OLIVIA          2      0.143
## 18    FLORA          2      0.143

```

```
event_scores
```

```

##   event_name raw_degree norm_degree
## 7          8         14      0.778
## 8          9         12      0.667
## 9          7         10      0.556

```

## 5	5	8	0.444
## 6	6	8	0.444
## 3	3	6	0.333
## 10	12	6	0.333
## 11	10	5	0.278
## 4	4	4	0.222
## 14	11	4	0.222
## 1	1	3	0.167
## 2	2	3	0.167
## 12	13	3	0.167
## 13	14	3	0.167

In a 2-mode network, the Normalized Degree Centrality represents the percentage of the opposing set that a node is connected to. It measures “participation rate” rather than just a raw count (Borgatti & Everett, 1997, p.254).

For women (participation rate): The normalized score represents the proportion of all events that a specific woman attended. For example, Evelyn has a raw degree of 8 and a normalized degree of 0.571. This means she attended 57.1% of all the events (8 out of 14). This tells us how active a woman is. A score of 1.0 would mean she attended every single event.

For events (participation rate): The normalized score represents the proportion of the total population of women that attended a specific event. For example, Event 8 has a raw degree of 14 and a normalized degree of 0.778. This means 77.8% of the entire group of women (14 out of 18) were present at this event. This tells us how popular an event is. A score of 1.0 would mean the all women showed up.

There are two main reasons why normalization is important here:

First, in a bipartite graph, a node cannot connect to others in its own set (e.g., a woman cannot “attend” another woman). Therefore, the theoretical maximum degree for any node is not the total network size minus one ($n-1$), but strictly the size of the opposing vertex set (Borgatti & Everett, 1997, p.254).

Second, we cannot compare scores across modes without normalization (because raw scores are on different scales, such as 18 women but only 14 events). But by dividing by the size of the opposite set (Woman’s degree / 14; Event’s degree / 18), we convert both to a standardized 0 to 1 scale. This allows us to compare the degree centrality of women and events. For instance, that Event 8 (0.778) is structurally more central to the network’s cohesion than the most active woman, Evelyn (0.571).

Based on the Normalized Degree Centrality results:

For women, Evelyn (0.571), Theresa (0.571), and Nora (0.571) have the highest normalized scores. This means they attended nearly 60% of all events, acting as the “glue” of the network. They are more likely bridge different social circles, ensuring that information flows across the entire group. However, Dorothy (0.143), Olivia (0.143) and Flora (0.143) have much lower scores, attending fewer than 15% of events. Structurally, these women are on the margins. They are less integrated into the overall structure and likely less active participants in this specific social circle.

For events, Event 8 (0.778) and Event 9 (0.667) are the structural peaks of the network. This means event 8 and 9 were attended by nearly 78% and 67% of all women, respectively. This exceptionally high score indicates a “mandatory” social activity that brought almost everyone together, regardless of their specific clique. Structurally, these events bridge the distance between individuals, allowing people from the “core” and “periphery” to interact. However, Event 1 (0.167), Event 2 (0.167), Event 13 (0.167), and Event 14 (0.167) were attended by only a small fraction of the women. This indicates that these events represent more specialized or exclusive gatherings, serving smaller subgroups rather than the network work.

Overall, the table shows that the highest normalized degree for a woman is 0.571 (not 1.0), even the most central women missed 40% of the events, so no single woman could dominate the entire network. In contrast, the highest normalized degree for an event is 0.778. The fact that singular events could capture nearly 80% of the population suggests that events are more cohesive than people.

3.2 “Woman by Woman” Centrality Measure

```
# woman x woman centrality
davis_projections <- bipartite_projection(davis)
davis_woman <- davis_projections$proj1

# raw and normalized degree
degree_raw <- degree(davis_woman, normalized = FALSE)
degree_norm <- degree(davis_woman, normalized = TRUE)

# table
woman_proj_scores <- data.frame(
  name = V(davis_woman)$name,
  raw_degree = degree_raw,
  norm_degree = round(degree_norm , 3)
)

# sort
woman_proj_scores <- woman_proj_scores[order(woman_proj_scores$norm_degree, decreasing = TRUE), ]
woman_proj_scores
```

##	name	raw_degree	norm_degree
##	EVELYN	17	1.000
##	THERESA	17	1.000
##	RUTH	17	1.000
##	VERNE	17	1.000
##	SYLVIA	17	1.000
##	NORA	17	1.000
##	HELEN	17	1.000
##	PEARL	16	0.941
##	MYRNA	16	0.941
##	KATHERINE	16	0.941
##	DOROTHY	16	0.941
##	LAURA	15	0.882
##	BRENDA	15	0.882
##	FRANCES	15	0.882
##	ELEANOR	15	0.882
##	OLIVIA	12	0.706
##	FLORA	12	0.706
##	CHARLOTTE	11	0.647

If a woman’s raw degree is 17, it means she shares at least one event with every other woman in the network. If her normalized degree is 1.000, it indicates that she is fully connected (she has a tie to every other woman).

```
# weight < 3, num_components = 4, too fragmented
net_threshold_3 <- delete_edges(davis_woman, E(davis_woman)[weight < 3])
num_components <- count_components(net_threshold_3)
num_components
```

```
## [1] 4
```

```
# weight < 2, num_components = 1, optimal
net_threshold_2 <- delete_edges(davis_woman, E(davis_woman)[weight < 2])
num_components <- count_components(net_threshold_2)
num_components
```

```
## [1] 1

# Threshold
davis_woman_strong <- delete_edges(davis_woman, E(davis_woman)[weight < 2])

# recompute centrality
degree_new_raw <- degree(davis_woman_strong, normalized = FALSE)
degree_new_norm <- degree(davis_woman_strong, normalized = TRUE)

# old norm degree
degree_old_norm <- degree(davis_woman, normalized = TRUE)

# comparison table
comparison_df <- data.frame(
  name = V(davis_woman_strong)$name,
  old_norm_degree = round(degree_old_norm, 3),
  new_norm_degree = round(degree_new_norm, 3),
  change = round(degree_new_norm - degree_old_norm, 3)
)

# sort
comparison_df <- comparison_df[order(comparison_df$new_norm_degree, decreasing = TRUE), ]
comparison_df
```

##		name	old_norm_degree	new_norm_degree	change
##	THERESA	THERESA	1.000	0.882	-0.118
##	RUTH	RUTH	1.000	0.882	-0.118
##	EVELYN	EVELYN	1.000	0.824	-0.176
##	NORA	NORA	1.000	0.824	-0.176
##	PEARL	PEARL	0.941	0.765	-0.176
##	VERNE	VERNE	1.000	0.765	-0.235
##	SYLVIA	SYLVIA	1.000	0.765	-0.235
##	LAURA	LAURA	0.882	0.706	-0.176
##	BRENDA	BRENDA	0.882	0.706	-0.176
##	ELEANOR	ELEANOR	0.882	0.706	-0.176
##	MYRNA	MYRNA	0.941	0.588	-0.353
##	KATHERINE	KATHERINE	0.941	0.588	-0.353
##	HELEN	HELEN	1.000	0.588	-0.412
##	FRANCES	FRANCES	0.882	0.471	-0.412
##	DOROTHY	DOROTHY	0.941	0.471	-0.471
##	CHARLOTTE	CHARLOTTE	0.647	0.412	-0.235
##	OLIVIA	OLIVIA	0.706	0.118	-0.588
##	FLORA	FLORA	0.706	0.118	-0.588

I mainly filtered out weak ties on the one-mode (Woman $\times$ Woman) projection. Specifically, I removed edges with a weight of 1 (single co-attendance) and kept only edges where women co-attended at least two events (weight ≥ 2). I choose `threshold < 2` because the network remained a single connected component: it removed noise (incidental contact) without destroying the structural integrity of the social group while when `threshold < 3`, the network broke into 4 disconnected components (too fragmented).

It is appropriate for this dataset because in the original one-mode projection, seven different women (Evelyn, Theresa, Ruth, Verne, Sylvia, Nora, Helen) all had a normalized degree of 1.000. This suggests they were all connected to everyone in the network. When nearly half the population has a “perfect” score, the centrality measure loses its analytical power to distinguish the “true” core from the slightly less active members. So by filtering out “one-off” meetings (weight < 2), I successfully increased the variance in the centrality scores, from 0.65 to 0.1 as shown in the table above, allowing for a meaningful ranking of the women.

As for down-weight co-attendance at larger events, I agree with that. As many researchers like Newman (2001) argue that, a tie formed by numerous individuals at a large-scale event (e.g., a university-wide job fair) is sociologically weaker than a tie formed by a small group in an intimate setting (e.g., a private dinner). For example, at a dinner party of 5 people, it is likely that everyone interacts with everyone else (strong ties) while at a university job fair of 500 people, an individual might only interact with a few people. However, the standard projection method treats all co-attendees as connected (because everyone present at job fair), which artificially inflates network density and centrality scores. Therefore, the links generated by large events are often “weak” or even “false” (false positives)(co-attendance $\neq$ interaction). In short, down-weighting such events is necessary.

However, it doesn’t support the strategy I chose for this specific Davis Southern Women network for two key reasons:

First, the “large” events in the Davis data (e.g., event 8 with 14 women) are still relatively small social gatherings in a small town context (Natchez, Mississippi). Unlike a massive job fair, co-attendance at an event of 14 likely implies meaningful face-to-face interaction. Down-weighting these events based on size alone would incorrectly treat them as sources of weak ties, potentially obscuring the very core of the social structure.

Second, my thresholding approach privileges recurrence over exclusivity. By filtering out “one-off” meetings (setting a threshold of weight ≥ 2), I assume that repeated interaction is a stronger indicator of a tie than a single meeting. This successfully solved the problem of “saturation” in the original data, where too many women (nearly half) had perfect scores. This increased the variance in centrality scores (from a 0.88–1.0 range to a 0.1–0.8 range), allowing for a clearer differentiation between core and peripheral members without fragmenting the network into disconnected components.

As for comparison of degree centrality scores across modes:

In the weighted one-mode projection, a high score indicates that a woman has co-attended at least one event with a large proportion of the other women. Seven women (including Evelyn, Theresa, and Nora) had a perfect score of 1.000, meaning they were connected to everyone. This lack of variation makes it difficult to distinguish the true core members from the rest, as the network appears overly dense due to large events creating connections between everyone present.

In contrast, the re-scaled version (where ties < 2 were removed) measures stronger social integration. By counting only relationships where women met at least twice, this measure filters out incidental contacts. The scores now reflect the breadth of a woman’s recurrent relationships. Consequently, the scores dropped significantly (e.g., Theresa dropped from 1.000 to 0.882, and Flora dropped from 0.706 to 0.118), revealing a clear hierarchy. This version successfully differentiates between women who are deeply embedded in the core social circle and those who are on the periphery.

Finally, these scores differ fundamentally from the original 2-mode centrality. In the 2-mode network, centrality measured participation (the percentage of events a woman attended). For example, Evelyn’s 2-mode score of 0.571 simply meant she was active. However, her 1-mode scores measure her connections to people, suggesting that after filtering out all weak ties (i.e., connections involving only a single co-attendance), Evelyn still maintains strong ties (co-attending at least two events) with 82.4% of the other women (14 women) in the network.

3.3 “Woman by Event” ERGM

Descriptive Measures for Two-Mode Networks

```
A <- as_incidence_matrix(davis, types = V(davis)$type)

num_women <- nrow(A)
num_events <- ncol(A)

cat("Women:", num_women, "Events:", num_events, "\n")
```

```
## Women: 18 Events: 14
cat("=== Woman Activity ===\n")

## === Woman Activity ===
table(rowSums(A))

##
## 2 3 4 5 6 7 8
## 3 1 6 1 1 3 3
summary(rowSums(A))

##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      2.000  4.000  4.000  4.944  7.000  8.000
cat("=== Event Popularity ===\n")

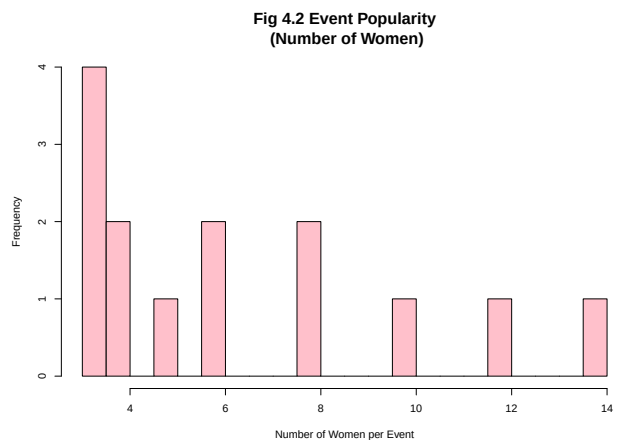
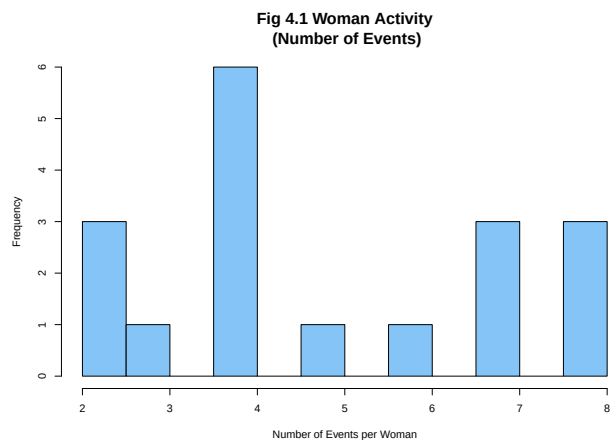
## === Event Popularity ===
table(colSums(A))

##
## 3 4 5 6 8 10 12 14
## 4 2 1 2 2 1 1 1
summary(colSums(A))

##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      3.000  3.250  5.500  6.357  8.000  14.000
par(mfrow = c(1, 2), cex.main = 1.5)

woman_activity <- rowSums(A)
hist(woman_activity, breaks = 20, col = "#84C4F6",
     main = "Fig 4.1 Woman Activity\n (Number of Events)",
     xlab = "Number of Events per Woman")

event_popularity <- colSums(A)
hist(event_popularity, breaks = 20, col = "pink",
     main = "Fig 4.2 Event Popularity\n (Number of Women)",
     xlab = "Number of Women per Event")
```



```
two_paths <- A %*% t(A)
diag(two_paths) <- NA
summary(two_paths)
```

```
##          EVELYN          LAURA          THERESA          BRENDA
## Min.      :1.000    Min.      :0.000    Min.      :1.000    Min.      :0.000
## 1st Qu.:2.000    1st Qu.:1.000    1st Qu.:2.000    1st Qu.:1.000
## Median :2.000    Median :2.000    Median :3.000    Median :2.000
## Mean     :2.941    Mean     :2.647    Mean     :3.353    Mean     :2.706
## 3rd Qu.:3.000    3rd Qu.:4.000    3rd Qu.:4.000    3rd Qu.:4.000
## Max.     :7.000    Max.     :6.000    Max.     :7.000    Max.     :6.000
## NA's     :1       NA's     :1       NA's     :1       NA's     :1
##          CHARLOTTE          FRANCES          ELEANOR          PEARL
## Min.      :0.000    Min.      :0.000    Min.      :0.000    Min.      :0.000
## 1st Qu.:0.000    1st Qu.:1.000    1st Qu.:1.000    1st Qu.:2.000
## Median :1.000    Median :1.000    Median :2.000    Median :2.000
## Mean     :1.412    Mean     :1.882    Mean     :2.118    Mean     :1.824
## 3rd Qu.:2.000    3rd Qu.:3.000    3rd Qu.:3.000    3rd Qu.:2.000
## Max.     :4.000    Max.     :4.000    Max.     :4.000    Max.     :3.000
## NA's     :1       NA's     :1       NA's     :1       NA's     :1
##          RUTH          VERNE          MYRNA          KATHERINE
## Min.      :1.000    Min.      :1.000    Min.      :0.000    Min.      :0.000
## 1st Qu.:2.000    1st Qu.:2.000    1st Qu.:1.000    1st Qu.:1.000
## Median :2.000    Median :2.000    Median :2.000    Median :2.000
## Mean     :2.353    Mean     :2.235    Mean     :1.941    Mean     :2.176
## 3rd Qu.:3.000    3rd Qu.:3.000    3rd Qu.:3.000    3rd Qu.:3.000
## Max.     :4.000    Max.     :4.000    Max.     :4.000    Max.     :6.000
## NA's     :1       NA's     :1       NA's     :1       NA's     :1
##          SYLVIA          NORA          HELEN          DOROTHY          OLIVIA
## Min.      :1.000    Min.      :1.000    Min.      :1       Min.      :0.000    Min.      :0.0000
## 1st Qu.:2.000    1st Qu.:2.000    1st Qu.:1       1st Qu.:1.000    1st Qu.:0.0000
## Median :2.000    Median :2.000    Median :2       Median :1.000    Median :1.0000
## Mean     :2.706    Mean     :2.529    Mean     :2       Mean     :1.412    Mean     :0.8235
## 3rd Qu.:4.000    3rd Qu.:3.000    3rd Qu.:3       3rd Qu.:2.000    3rd Qu.:1.0000
## Max.     :6.000    Max.     :6.000    Max.     :4       Max.     :2.000    Max.     :2.0000
## NA's     :1       NA's     :1       NA's     :1       NA's     :1       NA's     :1
##          FLORA
## Min.      :0.0000
## 1st Qu.:0.0000
## Median :1.0000
## Mean     :0.8235
## 3rd Qu.:1.0000
## Max.     :2.0000
## NA's     :1
```

ERGMs

```
davis_net <- as.network(A, bipartite = TRUE, directed = FALSE)
```

```
model_davis <- ergm(
  davis_net ~ edges +
    b1degree(c(2,4)) +
    b2degree(c(3,4)),
  control = control.ergm(MCMLE.maxit = 10)
)
```

```
summary(model_davis)
```

```
## Call:
## ergm(formula = davis_net ~ edges + b1degree(c(2, 4)) + b2degree(c(3,
##      4)), control = control.ergm(MCMLE.maxit = 10))
##
## Monte Carlo Maximum Likelihood Results:
##
##           Estimate Std. Error MCMC % z value Pr(>|z|)
## edges          0.2414    0.1789      0   1.350  0.17715
## b1degree2       2.7872    1.0206      0   2.731  0.00631 **
## b1degree4       1.6304    0.7109      0   2.294  0.02181 *
## b2degree3       4.3592    1.1109      0   3.924 < 1e-04 ***
## b2degree4       2.4789    1.1123      0   2.229  0.02584 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##      Null Deviance: 349.3  on 252  degrees of freedom
## Residual Deviance: 302.6  on 247  degrees of freedom
##
## AIC: 312.6  BIC: 330.2  (Smaller is better. MC Std. Err. = 0.1998)
```

In modeling the Davis Southern Women two-mode network, my aim to capture the probability of a tie (attendance) existing between a woman (Mode 1) and an event (Mode 2). The selected model formula includes three key terms:

First, **edges**: This is the foundational baseline term for any ERGM, functioning similarly to an intercept in a regression model. It captures the overall density of the network, representing the baseline tendency for women to attend events before considering any structural effects.

Second, **b1degree(c(2,4))**: These terms model the activity levels of the women (Mode 1). By specifying **c(2,4)**, I add parameters for women who attended exactly 2 or exactly 4 events(as seen in Fig 4.1 or `table(rowSums(A))` output, which shows a significant number of women falling into these specific activity counts). Including these specific counts allows the model to better fit the observed heterogeneity in social activity: some women are occasional participants, while others are “social hubs.”

Third, **b2degree(c(3,4))**: These terms model the popularity of the events (Mode 2). By specifying **c(3,4)**, I add parameters for events that had exactly 3 or 4 attendees(as seen in Fig 4.2 or `table(colSums(A))` output). This helps the model match the popularity distribution of events: some events attract small groups(small gatherings), while others attract larger crowds(large social events).

Together, these degree terms allow the model to move beyond a simple random graph by explicitly accounting for the fact that social participation is not uniform: specific levels of activity and popularity are structurally significant in this community.

Finally, **control = control.ergm(MCMLE.maxit = 10)** limits the fitting procedure to 10 iterations. While a full research analysis would require more iterations for convergence, this setting ensures the model runs quickly for exploratory purposes.

Several ERGM terms would not be appropriate for this dataset: bipartite networks have no triangles, no transitivity, no reciprocity, no clustering, and therefore one-mode ERGM terms such as `triangle`, `gwesp`, `gwdsp`, `mutual` are not valid for this dataset. Since the Davis dataset provides only attendance information and no attributes for women or events(e.g., age, race and marital status), attribute-based `nodematch` or homophily terms are also not appropriate here.

One challenge I ran into was deciding which ERGM terms were appropriate for capturing heterogeneity in women’s activity levels and event popularity. I initially experimented with more complex bipartite terms, such as `gwb1degree`, `gwb2degree` (to model the geometry-weighted degree distribution), as well as

b1concurrent and b2concurrent (to capture highly active women or highly popular events). Although these models ran successfully, the corresponding coefficients were consistently non-significant and did not improve the model fit. So in the end, I just went back to use `b1degree()` and `b2degree()` as tutorial suggested.

Interpretation:

In `model_davis`, I include the baseline edges term along with bipartite degree class terms for women (`b1degree`) and events (`b2degree`). This specification allows the model to capture the heterogeneity in participation patterns that I observed earlier in the descriptive analysis.

The edges coefficient is 0.24, which is not statistically significant ($p = 0.22$). This indicates that once we account for the specific degree distributions (the activity of women and the size of events), there is no significant baseline tendency for ties to form.

The `b1degree` (Women's Activity) terms capture the participation patterns of the women. The coefficient for `b1degree(2)` is 2.82 ($p < 0.01$) and `b1degree(4)` is 1.63 ($p < 0.05$). Both are positive and significant. This indicates a structural tendency for women to attend these specific activities: women attending exactly 2 or 4 events are significantly more frequent than expected by chance. This reflects specific subgroups of women who have consistent levels of social engagement.

The `b2degree` (Event Popularity) terms describe the size of the social gatherings. The `b2degree(3)` coefficient is the strongest effect in the model at 4.37 ($p < 0.001$), and `b2degree(4)` is 2.50 ($p < 0.05$). These large, positive values indicate a strong popularity mechanism: events in this society have a high popularity to attract groups of exactly 3 or 4 attendees. The network is structured heavily by these event sizes, suggesting that social gatherings tend to form in small and consistent events rather than varying randomly in size.

Overall, this model indicates that tie formation in the Davis Southern Women network is shaped by degree-based structural mechanisms on both sides of the bipartite graph. It is not a random collection of attendances; rather, it is structured by specific tendencies for women to attend certain events and for events to have specific appeal.

AI Disclosure Statement and Learning Reflections

I mainly used AI(ChatGPT) to help me with the degree centrality normalization part. Specifically, I applied the following `ifelse` statement: `V(davis)$norm_degree <- ifelse(V(davis)$type == FALSE, (V(davis)$degree / num_events), (V(davis)$degree / num_women))`. This ensures that women's degrees are divided by `num_events` while event degrees are divided by `num_women`.

But I am still not sure how to make these figures clearer(fig1-fig3), and I'm not fully confident that my approach for reducing edges to avoid the "ball of yarn" look is correct.

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